Research Statement

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Figure 1: Our illumination-aware generative models offer precise relighting of scenes without altering their geometry or material properties. With applications in computer graphics, virtual and augmented reality, and robotics, it enables realistic and convincing lighting effects for diverse environments.

Incorporating illumination properties into existing models has been a primary focus of my research. Traditional methods that decompose images into albedo and shading often lead to unrealistic results, such as the loss of surface details or changes in the spatial layout of the scene. To address these challenges, I have developed a novel image decomposition technique that modifies the traditional albedo-shading decomposition by separating persistent maps, analogous to albedo and encoding materials and sharp edges, from transient maps, analogous to shading and encoding diffuse and specular shading without geometry information (surface normals).

This technique is based on the Retinex theory [10] and involves the use of Mondrian with fine edges to encode persistent maps and smooth Perlin noise containing white blobs to encode transient maps. The use of these specific patterns allows for the separation of image information into light-dependent and light-independent components, enabling the realistic manipulation of lighting in images and scenes without altering their albedo or geometry. By imposing decomposition consistency on image parts, this technique allows for the successful transfer of objects from one image to another without losing any surface details and enables the realistic relighting of complex scenes.

I have applied my novel image decomposition technique to develop two state-of-the-art (SOTA) object insertion methods: DIPR [6] and Illuminator [2]. Both methods are capable of abstracting scene lighting and decoding images with realistic novel object insertion into scenes, allowing for the successful transfer of objects from one image to another without losing any surface details. In addition to object insertion, I have also developed the first single image relighting method, StyLitGAN [3], which can generate realistic scenes with arbitrary lighting, including cast shadows, inter-reflections, and soft shadows without requiring any physical models or other CGI (computer-generated imagery) labeled data (Figure 1). This is a significant improvement over previous methods, which often require extensive physical modeling or rely on large amounts of labeled data to produce similar results [7]. StyLitGAN can achieve these results by preserving scene semantics, ensuring that the resulting images are not only photorealistic but also semantically accurate.

My research on enhancing AI capabilities with illumination properties has significant implications for a wide range of industries and applications. This technique can be applied to create new types of content, such as virtual advertisements that can be inserted into live video streams or dynamic lighting for video games. In the realm of augmented and virtual reality experiences, my research offers exciting possibilities for enhancing the user experience. By enabling more realistic and convincing virtual environments, my techniques can help to create more engaging and immersive experiences for users. This has implications for a wide range of applications, including CGI for film, video games, training simulations, education, and entertainment.

Future Research: Moving forward, my research will continue to focus on developing advanced models that can generate realistic and detailed graphics with faster rendering times and handle

complex scenes effortlessly. By integrating both generative and robust discriminative properties [9, 1, 5, 8] in a unified model, I aim to enable comprehensive scene understanding and content creation for a wide range of applications. Ultimately, my work will contribute to the development of more reliable and efficient AI systems that can handle challenging conditions and reduce the risk of failure in critical applications, making a significant impact on the safety and reliability of real-world technologies.

Milestones

To ensure that my objectives are achieved within the timeframe of the fellowship, I have identified several milestones for each objective. These milestones will help me track my progress and make necessary adjustments to my research plan as needed.

Objective 1: Improving Image-Based Lighting

Milestone 1: Develop targeted and controllable image relighting algorithm. This algorithm will extend my own research and will allow for precise and specific changes to the illumination of an image, which will be achieved through the development of novel techniques and approaches.
Milestone 2: Develop a robust material editing algorithm. I plan to develop a robust material editing algorithm that can handle complex material interactions and accurately predict material behavior under different lighting conditions. This milestone will be achieved by building on existing research and incorporating machine learning techniques to improve accuracy.

These two milestones will be achieved by the end of the first year of the fellowship and will be a significant step towards improving the robustness of AI systems in visual editing applications

Objective 2: Improving Robustness

Milestone 3: Design algorithms and test robustness improvements by incorporating illumination properties into existing state-of-the-art discriminative models such as depth and segmentation estimators that are affected by lighting variations. To achieve this milestone, I will devise and implement robust representation learning algorithms that are dependent on a parametric lighting model. This will involve setting up experiments to measure the performance of the techniques, collecting and analyzing data, and improving the techniques based on the outcomes. This milestone will also be achieved by the first half of the second year of the fellowship.

Objective 3: Build a General Purpose, Unified Generative-Discriminative Model

Milestone 4: Develop an All-in-One Model for Comprehensive Scene Understanding and Content Creation. This model will integrate the advancements made in the first two objectives and use a generative model as a backbone for content creation and scene property extraction. The model will be capable of extracting different scene properties like depth, normals and segmentations from the same generative model without training new modules [4]. This milestone will be achieved by the end of the second year of the fellowship.

External Needs

To achieve these objectives, I will require access to computing resources such as GPU clusters and high-performance computing systems. I will also require laboratory equipment for testing and implementing the techniques developed. I plan to meet these needs by collaborating with other researchers, applying for grants, and utilizing the resources available at the institution where I will conduct my research.

References

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